

Tables

Table 1: Descriptive Statistics	
Variable	Mean (SD)
Panel A: Educational Outcomes	
Enrolled in school (1=Yes, 0=No)	0.89 (0.31)
Years of education (Years)	2.49 (2.62)
Panel B: Treatment Variable	
Household has electricity (1=Yes, 0=No)	0.27 (0.45)
Panel C: Instrumental Variables	
Distance to hypothetical network - least-cost (km)	82.53 (120.92)
Distance to hypothetical network - Euclidean (km)	93.86 (121.48)
Panel D: Geographic Controls	
Elevation (meters)	1270.09 (707.56)
Slope (degrees)	3.31 (4.13)
Distance to major road (km)	6.13 (12.32)
Distance to major river (km)	14.60 (10.58)
Distance to nearest 1989 city (km)	109.21 (122.36)
Population of nearest 1989 city	202549.22 (258029.24)
Panel E: Individual Characteristics	
Age (years)	9.51 (2.86)
Female (1=Yes, 0=No)	0.50 (0.50)
Urban residence (1=Yes, 0=No)	0.28 (0.45)

Notes: The sample includes children aged 5–14 from Kenya DHS 2014 and 2022 (N = 90,080). Standard deviations in parentheses. All distance variables measured in kilometers. Population refers to 1989 urban center population from official Kenya census records. Binary variables take values 0 or 1. Instrumental Variables: Distance to hypothetical 1990 electricity network constructed by connecting pre-existing power generation facilities (Olkaria Geothermal, Masinga Dam, Kiambere Dam, Gitaru Dam, Kindaruma Dam) to major urban centers from the 1989 Population Census using minimum spanning tree algorithms. Least-cost distance accounts for terrain and river crossings; Euclidean distance is straight-line. Geographic Controls: Elevation and slope from NASA SRTM (~90m resolution via AWS). Distances to major roads are sourced from OpenStreetMap (Geofabrik Kenya, 2024). Distances to major rivers are sourced from HydroRIVERS v1.0 (Strahler order ≥4). Distances to 1989 cities and populations from Kenya Population and Housing Census 1989. Outcome Variables: School enrollment and years of education from DHS. Treatment variable: Household electricity access from DHS. Individual characteristics: Age is complete years, gender is binary where 1=Female and 0=Male and urban is a binary where 1=urban and 0 otherwise.

Table 2: First Stage - Log Distance to Network → Household Electricity

	(1)	(2)	(3)
Log(Distance Euclidean)	-0.097***	-0.097***	-0.070***
	(0.008)	(0.008)	(0.007)
Observations	90059	90059	89979
R-squared	0.205	0.205	0.271
Child Controls	No	Yes	Yes
Geography and Infrastructure Controls	No	No	Yes
Counties	47	47	47
County Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Mean of Dependent Variable	0.273	0.273	0.273
SD of Dependent Variable	0.445	0.445	0.445
First-Stage F-Statistic	149.04	149.08	88.65

Notes. Individual-level regressions for children aged 5-14 in Kenya, using DHS data from 2014 and 2022. Dependent variable: Household has electricity (binary indicator). Log(Distance Euclidean)=Natural logarithm of distance to hypothetical 1989 electricity transmission network(Euclidean). Child controls include age and gender (female indicator). Geographic controls include elevation (meters), slope (degrees), agro-ecological zone indicators, log distance to major roads, and log distance to major rivers. Progressive controls: (1) County FE + Year FE (2) adds child controls to the baseline; (3) adds geographical and infrastructure controls ; All specifications include county and year fixed effects. Standard errors clustered at the cluster level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 3: Second Stage - IV Estimates

	Enrollment	Years of education
Household has Electricity (IV)	-0.023	0.613***
	(0.051)	(0.201)
Observations	89969	89910
R-squared	0.245	0.622
Child Controls	Yes	Yes
Geography and Infrastructure Controls	Yes	Yes
Counties	47	47
County Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Mean of Dependent Variable	0.892	2.486
SD of Dependent Variable	0.31	2.622
First-Stage F-Statistic	88.7	88.7

Notes. Individual-level regressions for children aged 5-14 in Kenya, using DHS data from 2014 and 2022. Dependent variables: School enrollment (binary) in Column 1 and years of education (continuous) in Column 2. Instrument: Log Euclidean distance (km) from DHS cluster to nearest segment of hypothetical 1989 electricity transmission network. Child controls include age and gender (female indicator). Geographic and infrastructure controls include elevation (meters), slope (degrees), agro-ecological zone indicators, log distance to major roads, and log distance to major rivers. All specifications include county fixed effects and year fixed effects. Standard errors clustered at the cluster level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 4: Heterogeneity Analysis - Effect of Electrification by Gender

	Years of Education
Household has Electricity (IV)	0.183 (0.210)
Child is Female	-0.090** (0.038)
Household has Electricity × Child is Female	0.868*** (0.124)
Observations	89910
R-squared	0.622
Child Controls	Yes
Geography and Infrastructure Controls	Yes
Counties	47
County Fixed Effects	Yes
Year Fixed Effects	Yes
Mean of Dependent Variable	2.486
SD of Dependent Variable	2.622
First-Stage F-Statistic	44.5

Notes. IV regression with interaction term for children aged 5-14 in Kenya, using DHS data from 2014 and 2022. Dependent variable: Years of education (continuous). β_1 = Effect for boys (when Female = 0). β_2 = Baseline gender gap (girls vs boys when not electrified). β_3 = Interaction term (differential effect of electricity for girls). Effect for boys = β_1 ; Effect for girls = $\beta_1 + \beta_3$. Instrument: Log Euclidean distance to hypothetical 1990 electricity transmission network, interacted with female. Child controls include age and gender. Geographic and infrastructure controls include elevation, slope, agro-ecological zones, log distance to roads and rivers. All specifications include county and year fixed effects. Standard errors clustered at cluster level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table A2: Robustness - Excluding Major City Counties

	Enrollment	Years of education
Household Has Electricity (IV)	-0.096	1.096***
	(0.112)	(0.408)
Observations	78432	78379
R-squared	0.243	0.607
Child Controls	Yes	Yes
Geography and Infrastructure Controls	Yes	Yes
Counties	39	39
County Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Mean of Dependent Variable	0.879	2.418
SD of Dependent Variable	0.327	2.601
First-Stage F-Statistic	28.3	28.3

Notes. Robustness check excluding 8 major city counties (Nairobi, Mombasa, Kisumu, Nakuru, Kiambu, Uasin Gishu, Machakos, Nyeri). Individual-level regressions for children aged 5-14 in Kenya, using DHS data from 2014 and 2022. Dependent variables: School enrollment (binary) in Column 1 and years of education (continuous) in Column 2. Instrument: Log Euclidean distance (km) from DHS cluster to nearest segment of hypothetical 1990 electricity transmission network. Child controls include age and gender. Geographic and infrastructure controls include elevation, slope, agro-ecological zones, log distance to roads and rivers. All specifications include county and year fixed effects. Standard errors clustered at cluster level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table A3: Robustness - Least-Cost Distance as Instrument

	Enrollment	Years of education
Household Has Electricity (IV)	-0.021	0.472
	(0.089)	(0.361)
Observations	89969	89910
R-squared	0.245	0.622
Child Controls	Yes	Yes
Geography and Infrastructure Controls	Yes	Yes
Counties	47	47
County Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Mean of Dependent Variable	0.892	2.486
SD of Dependent Variable	0.31	2.622
First-Stage F-Statistic	29.6	29.6

Notes. Robustness check using least-cost distance as alternative instrument. Individual-level regressions for children aged 5-14 in Kenya, using DHS data from 2014 and 2022. Dependent variables: School enrollment (binary) in Column 1 and years of education (continuous) in Column 2. Instrument: Log least-cost distance (km) accounting for terrain (elevation, slope) and river crossings to hypothetical 1990 electricity transmission network. Child controls include age and gender. Geographic and infrastructure controls include elevation, slope, agro-ecological zones, log distance to roads and rivers. All specifications include county and year fixed effects. Standard errors clustered at cluster level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$